黄 东 浩

+1 (571) XXXX-XXXX ｜donghao.huang@gmail.com｜美国弗吉尼亚州阿灵顿
新加坡公民｜专利：10 项已授权，50+项审批中

职业概述

拥有 26 年以上技术与管理经验的科技高管，覆盖软件工程、Web3、AI/ML/GenAI 以及支付创新领域。现任万事达卡（Mastercard）新兴技术方向（AI/Web3）全球研发负责人，同时攻读生成式 AI 工程博士。已发表 24 项研究成果，涵盖大语言模型（LLM）部署、LLM/ML 模型优化与应用，以及多智能体系统。具备将学术研究转化为金融服务领域企业级落地成果的成熟实践与卓越业绩。

教育背景

生成式人工智能工程博士 (Doctor of Engineering) 2023 年 8 月 – 2026 年 5 月

新加坡管理大学，新加坡（在读）

- 2025 年获新加坡资讯通信媒体发展局（IMDA）数字奖学金（研究生）
- 研究方向：检索增强生成（RAG）、边缘 AI、LLM 和 ML 模型优化、多智能体系统、情感分析、推理型 LLM
- 在 AI 学术会议和期刊发表 24 篇同行评审论文（AAAI、AAMAS、NAACL、IJCNN、PAKDD、IEEE CAI、IEEE SSCI、IEEE ICDM、IEEE Intelligent Systems、Artificial Intelligence Reviews）

软件工程技术硕士 2004 年 1 月 – 2008 年 1 月

新加坡国立大学，新加坡

新加坡经济发展局(EDB)研究与培训项目证书 2001 年 9 月 – 2002 年 8 月

新加坡国立大学光电子中心，新加坡

理论物理学硕士 1998 年 9 月 – 2001 年 7 月

北京大学，中国北京

- 在物理学术期刊发表 2 篇同行评审论文

工程物理学学士 1989 年 9 月 – 1995 年 7 月

清华大学，中国北京

职业经历

研发副总裁 / 全球新兴技术负责人 – Mastercard（美国） 2025 年 8 月 – 至今

美国弗吉尼亚州阿灵顿

- 领导 AI 新兴技术（EmTech）领域的全球研发计划
- 领导 5 个战略性"大赌注"项目的 EmTech 研究——旨在塑造 Mastercard 下一代支付基础设施和 AI 驱动商业解决方案的高影响力机密项目
- 组织并主持 SENTIRE'25（第 15 届情感分析与语言关联数据研讨会），与 IEEE ICDM 2025 联合举办，2025 年 11 月 13 日，美国华盛顿特区

研发副总裁 / 全球新兴技术负责人 – Mastercard 亚太 2022 年 5 月 – 2025 年 7 月

新加坡

- 领导新兴技术（EmTech）领域的全球研发计划：AI / 机器学习 / 生成式 AI 和 Web3
- AI/LLM/GenAI 项目：
  - 领导 Mastercard 助手平台（MAP）的 EmTech 研究，这是一套旨在简化 Mastercard 产品客户入门流程的数字助手
    - 使用 LangChain、FAISS 向量数据库和 HuggingFace 嵌入构建初始 RAG 原型
    - 在 NVIDIA A40 和 L40 GPU 上对开源 LLM（Llama 2、Orca 2）进行早期基准测试，展示 24-49%的推理速度提升
    - 开发创新的重复感知性能（RAP）评估框架，现已部署于 MAP 生产环境
    - 领导使用 Ollama 在多种硬件平台上进行边缘感知 LLM 部署研究
  - 使用 LLM（GPT-4/5、Claude、Llama、Qwen）开创企业 GenAI 应用，实现支付/商业自动化
    - 使用 QLoRA 在多 GPU 配置上微调 70B+参数模型，用于中英翻译和逻辑推理
    - 开发多智能体系统，使用边缘部署 LLM 实现 99.9%准确率的发票对账
    - 开发 HMASP——首个实现端到端智能体支付工作流的基于 LLM 的多智能体系统
    - 开发 CMAS4G2 协作多智能体系统，用于自动生成 Gherkin 验收测试
    - 创建 LLM-as-a-Judge 评估框架，相比 GPT-5 实现 47 倍成本降低
- Web3 项目：
  - 开发 Mastercard 代币化卡片 NFT 解决方案，支持将支付卡作为灵魂绑定代币（SBT）进行代币化
  - 开发 DigiPay，利用 Layer 2 区块链和 ERC-4337 账户抽象的 Web3 微支付稳定币解决方案

研发副总裁 / 新加坡研发负责人 – Mastercard 亚太 2011 年 8 月 – 2022 年 5 月

新加坡

- 创建并领导新加坡研发团队——Mastercard 全球创新与研发部门；管理位于新加坡的高绩效研发团队
- 从事支付/商业领域的创新与研发，涵盖前沿技术：EMV 芯片、物联网、二维码支付、车联网、人形机器人、对话式 AI、AR/VR、区块链和数字资产、Web3、NFT、DeFi、元宇宙
- 推动复杂项目从创意到商业化交接的全流程；促进项目向 Mastercard 商业化流程的过渡

- 作为技术创新的倡导者，与各类合作伙伴（内部部门、外部合作伙伴、大学）协作，确保 Mastercard 在支付领域建立技术卓越的声誉
- 主要商业化项目：
    - 2021 年 11 月：Mastercard 与新奥尔良市和 MoCaFi 合作推出 Crescent City Card 项目
    - 2020 年 9 月：Mastercard 与三星、Airtel 和 Asante 合作，通过按需付费服务推动非洲数字普惠
    - 2018 年 10 月：Mastercard、Centenary Bank 和 M-KOPA Solar 在乌干达推出首创能源解决方案
    - 2018 年 2 月：Mastercard 使用 Facebook Messenger 帮助小企业数字化
    - 2017 年 9 月：SPH Buzz 与 Mastercard 合作推出新型混合商店概念
    - 2017 年 2 月：BharatQR——全球首个可互操作的二维码受理解决方案在印度推出
    - 2016 年 5 月：MasterCard 为软银机器人人形机器人"Pepper"提供首个商业应用
    - 2015 年 9 月：MasterCard 变革援助物资分发方式
    - 2012 年 8 月：StarHub 与 DBS、EZ-Link 和 MasterCard 合作推出智能钱包

高级软件工程师 – PayPal 2008 年 7 月 – 2011 年 8 月

新加坡

- 领导软件开发团队，负责一系列针对巴西市场的项目，以满足这一快速增长市场的需求
- 限额/验证/合规领域的主题专家
- 主要商业化项目：
    - 领导 12 人开发团队，在 8 个月内交付最大的欧盟合规项目（EU KYB Receiving Limit）
    - 构建中国银联（CUP）集成，实现实时交易处理

软件研发经理 – Streaming21（新加坡） 2007 年 10 月 – 2008 年 7 月

新加坡

- 建立研发中心并领导媒体中继服务器的开发——一款领先的 IPTV 流媒体服务器

软件开发负责人 – Schmidt Electronics (S.E.A.) 2006 年 1 月 – 2007 年 9 月

新加坡

软件工程师 – Unaxis 新加坡 2005 年 2 月 – 2005 年 12 月

新加坡

软件工程师 – ASM Technology 新加坡 2002 年 9 月 – 2005 年 2 月

新加坡

软件工程师 – 北京智达科技 1995 年 8 月 – 1998 年 8 月

中国北京

志愿服务经历

会长 – 宝严（新加坡）教育协会 2025 年 11 月 – 至今

副会长 – 北京大学新加坡校友会 2016 年 6 月 – 2020 年 6 月

导师 – Startupbootcamp 金融科技新加坡 2015 年 1 月 – 2019 年 1 月

IT 与支付顾问 – ConneXions 国际（新加坡） 2011 年 3 月 – 2015 年 12 月

技术顾问 – 圣安德烈自闭症中心（新加坡） 2006 年 11 月 – 2008 年 7 月

技术技能

- AI/ML/GenAI：LLM（GPT-4/5、Claude、Gemini、Llama、Qwen、DeepSeek-R1、Mistral、Magistral、Phi、Nemotron）、PyTorch、Transformers、PEFT、LoRA/QLoRA 微调、LlamaFactory、Unsloth、RAG、LangChain、LangGraph、Autogen、Ollama、LM Studio、FAISS、向量数据库、少样本学习、微调、提示工程、情感分析、NLP
- GPU/硬件：NVIDIA H100、L40、A40、RTX A6000、RTX 4080、RTX 4090、Jetson AGX Orin、DGX Spark；Apple M 系列；4 位量化、模型优化、边缘 AI 部署
- 编程语言与框架：Python、C/C++、Java、NodeJS、React、Vue、Spring Boot/Cloud、Solidity
- 云与基础设施：AWS、Azure、Heroku；微服务、CI/CD、Docker
- Web3/区块链：区块链、IPFS、NFT、DeFi、x402 协议

专利

（10 项已授权，50+项待审 — 均归属万事达卡）

已授权专利

- **US12417452B2** – Smart chip payment acceptance (2025)
- **US12217246B2** – Method and system for use of an EMV card in a multi-signature wallet for cryptocurrency transactions (2025)
- **US11715100B2** – Electronic system and computerized method for verification of transacting parties to process transactions (2023)
- **US11615406B2** – Method and system for providing a service at a self-service machine (2023)
- **US10535034B2** – Item delivery management systems and methods (2020)
- **US10482499B2** – Method for conducting a transaction (2019)
- **US10423949B2** – Vending machine transactions (2019)
- **US10083427B2** – Method for receiving an electronic receipt of an electronic payment transaction into a mobile device (2018)
- **US9836729B2** – Method and system for conducting a payment transaction and corresponding devices (2017)
- **US8712914B2** – Method and system for facilitating micropayments in a financial transaction system (2014)

部分待审申请

- **WO2026063867A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **US20260080407A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **WO2024058723A1** – Digitization of payment cards for web 3.0 and metaverse transactions (2022)

- **US20230059546A1** – Access control system (2021)
- **EP4356332A2** – Method and system for mediated cross ledger stable coin atomic swaps using hashlocks (2021)
- **US20210406887A1** – Method and system for merchant acceptance of cryptocurrency via payment rails (2020)
- **US20210158316A1** – Electronic system and computerized method for controlling operation of service devices (2019)
- **US20200175489A1** – Method and system for QR code originated vending (2017)
- **WO2017200643A1** – Method and system for allocating and transacting with multiple non-financial commodities (2016)
- **US20180025332A1** – Transaction facilitation (2016)

... 以及 40+项其他待审专利申请，涵盖支付技术、区块链、AI 和物联网领域。

精选论文

1. Chua, J., 等. (2026). 一种新型基于大语言模型的层级多智能体支付系统. PAKDD 2026.
2. Huang, D., 等. (2026). 任务复杂度的重要性：大语言模型情感分析推理的实证研究. PAKDD 2026.
3. Huang, D., & Wang, Z. (2025). 使用 DeepSeek-R1 的可解释情感分析. IEEE Intelligent Systems.
4. Huang, D., 等. (2025). 边缘 LLM：性能与效率评估. IJCNN 2025.
5. Huang, D., 等. (2025). RAP：平衡重复与性能的指标. NAACL 2025.
6. Wang, Z., Huang, D., 等. (2025). 中文情感分析综述：主题、方法与趋势. Artificial Intelligence Review.
7. Lin, W. B., Huang, D. H., 等. (2001). 弱相互作用大质量粒子的非热产生与亚星系结构. Physical Review Letters.

... 及其它 19 篇同行评审论文（人工智能领域共 24 篇，物理领域共 2 篇）

DONGHAO HUANG

+1 (571) XXXX-XXXX | donghao.huang@gmail.com | Arlington, VA, USA
Singapore Citizen | Patents: 10 granted, 50+ pending

PROFESSIONAL SUMMARY

Technology executive with 26+ years of experience across software engineering, Web3, AI/ML/GenAI, and payments innovation. Currently leading global R&D for Mastercard Emerging Technology (AI/Web3) while pursuing a Doctor of Engineering in Generative AI. Author of 24 published research works spanning LLM deployment, LLM/ML model optimization and applications, and multi-agent systems. Proven track record of translating academic research into enterprise-scale financial services solutions.

EDUCATION

Aug 2023 – May 2026 (ongoing) Singapore Management University Singapore
Doctor of Engineering (EngD) in Generative AI

- Awarded IMDA SG Digital Scholarship (Postgraduate) 2025
- Research focus: Retrieval-Augmented Generation (RAG), Edge AI, LLM and ML model optimization, multi-agent systems, sentiment analysis, reasoning LLMs
- Published 24 peer-reviewed papers at AI conferences and journals (AAAI, AAMAS, NAACL, IJCNN, PAKDD, IEEE CAI, IEEE SSCI, IEEE ICDM, IEEE Intelligent Systems, Artificial Intelligence Reviews)

Jan 2004 – Jan 2008 National University of Singapore Singapore
Master of Technology in Software Engineering

Sep 2001 – Aug 2002 Centre for Optoelectronics, NUS Singapore
Certificate, EDB Research and Training Programme

Sep 1998 – Jul 2001 Peking University Beijing, China
Master of Science in Theoretical Physics

- Published 2 peer-reviewed papers in physics journals

Sep 1989 – Jul 1995 Tsinghua University Beijing, China
Bachelor of Engineering in Engineering Physics

PROFESSIONAL EXPERIENCE

Aug 2025 – to date Mastercard (U.S.) Arlington, VA, USA
VP, R&D / Global EmTech Lead

- Lead global R&D initiatives in AI Emerging Technology (EmTech) domains
- Lead EmTech research for 5 strategic "Big Bets" projects – high-impact, confidential initiatives aimed at shaping Mastercard's next-generation payment infrastructure and AI-powered commerce solutions
- Organized and hosted SENTIRE'25 (15th Workshop on Sentiment Analysis and Linguistic Linked Data) co-located with IEEE ICDM 2025, November 13th, Washington DC.

May 2022 – Jul 2025 Mastercard Asia/Pacific Pte Ltd Singapore

VP, R&D / Global EmTech Lead

- Lead global R&D initiatives in Emerging Technology (EmTech) domains: AI / Machine Learning / Generative AI and Web3.
- **AI/LLM/GenAI Initiatives:**
  - Led EmTech research for [Mastercard Assistant Platform]() (MAP)**,** a suite of digital assistants designed to simplify the customer onboarding process for Mastercard products
    - ➢ Built the initial Retrieval-Augmented Generation (RAG) prototype using LangChain, FAISS vector databases, and HuggingFace embeddings to power enterprise-grade knowledge systems.
    - ➢ Conducted early benchmarking of open-source LLMs (Llama 2, Orca 2) on NVIDIA A40 and L40 GPUs (48GB each), demonstrating 24-49% inference speed improvement on L40 vs A40.
    - ➢ Developed a novel Repetition-aware Performance (RAP) evaluation framework for LLM quality assessment, now fully deployed in the MAP production environment.
    - ➢ Led research on edge-aware LLM deployment using the Ollama framework across diverse hardware platforms (NVIDIA RTX A6000, NVIDIA RTX 4090, NVIDIA Jetson AGX Orin, Apple M-series—directly informing MAP's deployment strategy.
  - Pioneered enterprise GenAI applications using LLMs (GPT-4/5, Claude, Llama, Qwen) for payments/commerce automation
    - ➢ Fine-tuned 70B+ parameter models (Llama-3.1-70B, Qwen2-72B) using QLoRA on multi-GPU setups (4× NVIDIA L40 GPUs) for Chinese-English translation and logical reasoning
    - ➢ Developed multi-agent systems for invoice reconciliation achieving 99.9% accuracy using edge-deployed LLMs (qwen2.5:32b)
    - ➢ Developed hierarchy multi-agent systems for conversational payment agent (HMASP) - first LLM-based multi-agent system to implement end-to-end agentic payment workflows, enabling external AI agents to converse and complete payments using natural language with minimal integration overhead
    - ➢ Developed CMAS4G2, a cooperative multi-agent system for automated Gherkin acceptance test generation using Microsoft's AutoGen Swarm framework; demonstrated open-weight models on consumer hardware (NVIDIA RTX 4090, 24GB VRAM) approach proprietary performance (GPT-OSS 20B: 87.6% success vs Claude Sonnet 4: 90.4%)
    - ➢ Created LLM-as-a-Judge (LAJ) evaluation framework achieving 47× cost reduction vs GPT-5 with negligible accuracy loss for automated test coverage assessment
    - ➢ Evaluated reasoning LLMs (DeepSeek-R1, Qwen3, GPT-OSS) for sentiment analysis with cross-platform performance benchmarking on NVIDIA H100, NVIDIA GeForce RTX 4090, Apple M3 Max
    - ➢ Developed energy efficiency metrics (Energy per Inference, Performance-to-Power Ratio) for sustainable AI deployment
- **Web3 Initiatives:**
  - Developed Mastercard Tokenized Card NFT solution, enabling seamless tokenization of payment cards as Soul Bound Tokens (SBTs) into Web3 wallets
    - ➢ Enables users to store credit cards securely as non-transferable NFTs, providing publicly verifiable digital identity for web3 wallet addresses

- Integrates with crypto exchanges, merchant partners, and payment acquirers through smart contract architecture
- Simplifies Web3 user onboarding by eliminating the need to switch between multiple applications and cumbersome digital wallet funding processes

- Developed DigiPay, a Web3 micropayment stablecoin solution leveraging Layer 2 blockchain and ERC-4337 account abstraction
  - Implemented Digicents smart contract token with high TPS, low-latency blockchain for fast, efficient, and affordable transactions
  - Built smart contract wallet accounts with auto-top-up and auto-withdrawal features, eliminating manual on-ramp/off-ramp processes
  - Integrated with Tokenized Card NFT for secure funding and transaction authentication within the Digicents ecosystem

Aug 2011 – May 2022 Mastercard Asia/Pacific Pte Ltd Singapore

VP, R&D / Head of Singapore R&D

- Found and lead Singapore R&D team in Mastercard Foundry (previously known as Mastercard Labs) – Mastercard's global innovation and R&D division; managing a high performing R&D team based in Singapore.
- Work on innovation and R&D in payment/commerce domains across lots of latest and cutting-edge technologies, including but not limited to, EMV Chip technologies, Internet of Things, QR Payments, Connected Car, Humanoid Robots, Conversational AI, AR/VR, Blockchain and Digital Assets, Web3, NFT, DeFi, Metaverse and so on
- Drive complex initiatives from idea through to commercialization hand-over; facilitate the transition of initiatives into Mastercard commercialization process
- As a champion for Technological Innovation, collaborate with various partners (internal departments, external partners, universities) to ensure Mastercard attains a reputation for technological excellence in the Payments space
- **Key Commercialized Projects**
  - Nov 2021: Mastercard partners with city of New Orleans and MoCaFi to announce Crescent City Card program: https://www.mastercard.com/news/press/2021/november/mastercard-partners-with-city-of-new-orleans-and-mocafi-to-announce-crescent-city-card-program/
  - Sep 2020: Mastercard partners with Samsung, Airtel and Asante to drive digital inclusion in Africa through Pay-on-Demand services: https://newsroom.mastercard.com/mea/press-releases/mastercard-partners-with-samsung-airtel-and-asante-to-drive-digital-inclusion-in-africa-through-pay-on-demand-services/
  - Oct 2018: Mastercard, Centenary Bank and M-KOPA Solar roll-out first of its kind energy solution in Uganda: https://newsroom.mastercard.com/mea/press-releases/mastercard-centenary-bank-and-m-kopa-solar-roll-out-first-of-its-kind-energy-solution-in-uganda/
  - Feb 2018: Mastercard Uses Facebook Messenger To Help Small Businesses Go Digital: https://newsroom.mastercard.com/press-releases/mastercard-uses-facebook-messenger-help-small-businesses-go-digital/
  - Sep 2017: SPH Buzz Partners With Mastercard To Launch New Hybrid Store Concept: https://newsroom.mastercard.com/asia-pacific/press-releases/sph-buzz-partners-with-mastercard-to-launch-new-hybrid-store-concept/

- Feb 2017: BharatQR – World's first interoperable QR code acceptance solution launched in India: https://newsroom.mastercard.com/asia-pacific/press-releases/bharatqr-worlds-first-interoperable-qr-code-acceptance-solution-launched-in-india/
- May 2016: MasterCard Powers First Commerce Application within SoftBank Robotics' Humanoid Robot "Pepper": https://newsroom.mastercard.com/press-releases/mastercard-powers-first-commerce-application-within-softbank-robotics-humanoid-robot-pepper/
- Sep 2015: MasterCard Transforms Aid Distribution: https://newsroom.mastercard.com/press-releases/mastercard-transforms-aid-distribution/
- Aug 2012: StarHub unveils 'SmartWallet', in collaboration with DBS, EZ-Link and MasterCard: https://www.starhub.com/about-us/newsroom/2012/august/starhub-unveils-smartwallet--in-collaboration-with-dbs--ez-link-.html

Jul 2008 – Aug 2011 PayPal Singapore Development Centre Singapore
Senior Software Engineer / Team Lead

- Lead a team of software developers, working on a pipeline of projects related to Brazil to cater to this fast growing market
- Subject Matter Expert in limits / verification / compliance domain
- **Key Commercialized Projects**
  - Led 12-developer team delivering largest EU compliance project (EU KYB Receiving Limit) in 8 months
  - Built China UnionPay (CUP) integration enabling real-time transaction processing

Oct 2007 – Jul 2008 Streaming21 (Singapore) Pte Ltd Singapore
Software R&D Manager

- Established an R&D centre and led development of Media Relay Server, a leading IPTV streaming server.

Jan 2006 – Sep 2007 Schmidt Electronics (S.E.A.) Pte Ltd Singapore
Leader, Software Development

Feb 2005 – Dec 2005 Unaxis Singapore Pte Ltd Singapore
Software Engineer

Sep 2002 – Feb 2005 ASM Technology Singapore Pte Ltd Singapore
Software Engineer

Aug 1995 – Aug 1998 Beijing Zhida Technology Pte Ltd Beijing, China
Software Engineer

VOLUNTEERING EXPERIENCE

Nov 2025 – to date Bao Yan (Singapore) Education Society
President

Jun 2016 – Jun 2020 Peking University Alumni Association (Singapore)
Vice President

Jan 2015 – Jan 2019 Startupbootcamp FinTech Singapore
Mentor

Mar 2011 – Dec 2015 ConneXions International Singapore
IT and Payment Consultant

Nov 2006 – Jul 2008 St. Andrew's Autism Centre Singapore
Technology Consultant

TECHNICAL SKILLS

- **AI/ML/GenAI:** LLMs (GPT-4, GPT-5, Claude, Gemini, Llama, Qwen, DeepSeek-R1, Mistral, Magistral, Phi, Nemotron), PyTorch, Transformers, LLM/ML Model Fine-tuning and Optimization, PEFT, LoRA/QLoRA Fine-tuning, LlamaFactory, Unsloth, RAG, LangChain, LangGraph, Autogen, Ollama, LM Studio, Vector Databases, FAISS, Chroma, Prompt Engineering, Few-shot Learning, NLP, Sentiment Analysis, Agentic AI, OpenClaw, NemoClaw, Edge AI Deployment, Energy-efficient Inference
- **GPU/Hardware:** NVIDIA H100, L40, A40, RTX 4080, RTX 4090, RTX A6000, Jetson AGX Orin, DGX Spark; Apple M-series
- **Languages & Frameworks:** Python, C/C++, Java, NodeJS, React, Vue, Spring Boot/Cloud, Solidity
- **Cloud & Infrastructure:** AWS, Azure, Heroku; Microservices, CI/CD, Docker
- **Web3/Blockchain:** Blockchain, Smart Contract, IPFS, NFT, DeFi

PATENTS

(10 granted, 50+ pending — all assigned to Mastercard)

Granted Patents

- **US12417452B2** – Smart chip payment acceptance (2025)
- **US12217246B2** – Method and system for use of an EMV card in a multi-signature wallet for cryptocurrency transactions (2025)
- **US11715100B2** – Electronic system and computerized method for verification of transacting parties to process transactions (2023)
- **US11615406B2** – Method and system for providing a service at a self-service machine (2023)
- **US10535034B2** – Item delivery management systems and methods (2020)
- **US10482499B2** – Method for conducting a transaction (2019)
- **US10423949B2** – Vending machine transactions (2019)
- **US10083427B2** – Method for receiving an electronic receipt of an electronic payment transaction into a mobile device (2018)
- **US9836729B2** – Method and system for conducting a payment transaction and corresponding devices (2017)
- **US8712914B2** – Method and system for facilitating micropayments in a financial transaction system (2014)

Selected Pending Applications

- **WO2026063867A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **US20260080407A1** – Method and system for seamless cross-network physical-digital (phy-gital) experience (2024)
- **WO2024058723A1** – Digitization of payment cards for web 3.0 and metaverse transactions (2022)
- **US20230059546A1** – Access control system (2021)
- **EP4356332A2** – Method and system for mediated cross ledger stable coin atomic swaps using hashlocks (2021)
- **US20210406887A1** – Method and system for merchant acceptance of cryptocurrency via payment rails (2020)

Mr HUANG, DONGHAO (黄东浩) +1 (571) XXXX-XXXX donghao.huang@gmail.com

- **US20210158316A1** – Electronic system and computerized method for controlling operation of service devices (2019)
- **US20200175489A1** – Method and system for QR code originated vending (2017)
- **WO2017200643A1** – Method and system for allocating and transacting with multiple non-financial commodities (2016)
- **US20180025332A1** – Transaction facilitation (2016)

… *and 40+ additional pending patent applications across payment technology, blockchain, AI, and IoT domains.*

RESEARCH PAPERS

(papers at top-tier venues are highlighted in yellow; full texts are accessible via the hyperlinked paper titles)

Artificial Intelligence

1. Chua, J., Huang, D. & Wang, Z. (2026). A Novel Hierarchical Multi-Agent System for Payments Using LLMs. Accepted for publication in *The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026).*
2. Huang, D., & Wang, Z. (2026). Task Complexity Matters: An Empirical Study of Reasoning in LLMs for Sentiment Analysis. Accepted for publication in *The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026).*
3. Huang, D., Chua, J. K., & Wang, Z. (2026). Beyond Task Success: Measuring Workflow Fidelity in LLM-Based Agentic Payment Systems. Accepted for publication in *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2026.*
4. Huang, D., Pai, P., & Wang, Z. (2026). Entity Matching with LLMs at Scale: Accuracy, Cost, and Reasoning Effects. Accepted for publication in *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2026.*
5. Huang, D., Chew, S., & Wang, Z. (2026). CMAS4G2: A Locally Deployed Cooperative Multi-Agent System for Gherkin Acceptance Test Generation. Accepted for publication in *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2026.*
6. Huang, D. & Pandey, M. (2026). An Agentic Voice-Based Assistant for Interactive Conversation and Guidance in Real-World Environments. Accepted for publication in Demonstration Track, *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*, Paphos, Cyprus, May 25–29, 2026.
7. Chua, J. K., Tse, C. H., Huang, D., & Wang, Z. (2026). ConvPayMAS: Conversational Payment Multi-Agent System with Agent-to-Agent Protocol and Three-Mandate Verification. Accepted for publication in Demonstration Track, *Proc. of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026)*, Paphos, Cyprus, May 25–29, 2026.
8. Huang, D., Malwe, G., & Wang, Z. (2026). When Agents Fail to Act: A Diagnostic Framework for Tool Invocation Reliability in Multi-Agent LLM Systems. Accepted for publication in *2026 The 9th International Conference on Artificial Intelligence and Big Data (ICAIBD 2026).*
9. Huang, D., Chew S., Dutkiewicz A., & Wang, Z. (2026). LLM-as-a-Judge for Scalable Test Coverage Evaluation: Accuracy, Operational Reliability, and Cost. Presented at *AAAI 2026 Workshop on Next-Gen Code Development with Collaborative AI Agents.*
10. D. Huang, N. Teo, A. Chekuru, E. Deutschman and Z. Wang, "Deploying Reasoning LLMs for Sentiment Analysis: Architecture Trade-offs on Consumer Hardware," 2025 IEEE International Conference on Data Mining Workshops (ICDMW), Washington, DC, USA, 2025, pp. 2048-2055, doi: 10.1109/ICDMW69685.2025.00249.
11. S. Samuel, D. Huang, J. T. Y. Hong and Z. Wang, "Integrating LLM Sentiment Analysis into Machine Learning for Soccer Betting," 2025 IEEE International Conference on Data Mining Workshops (ICDMW), Washington, DC, USA, 2025, pp. 2156-2164, doi: 10.1109/ICDMW69685.2025.00262.
12. D. Huang and Z. Wang, "Explainable Sentiment Analysis With DeepSeek-R1: Performance, Efficiency, and Few-Shot Learning," in *IEEE Intelligent Systems*, vol. 40, no. 6, pp. 52-63, Nov.-Dec. 2025, doi: 10.1109/MIS.2025.3614967.
13. Huang, D., Ng, D., Pen, H., Wang, Z., & Cambria E. (2025). Evaluating the Impact of LLM-Manipulated Content on Fake News Detection. In: Yuan, S., Malliaros, F., Zheng, X. (eds) *Trends and Applications in Knowledge*

Discovery and Data Mining. PAKDD 2025. Lecture Notes in Computer Science(), vol 15835. Springer, Singapore. https://doi.org/10.1007/978-981-96-8197-6_28

14. Teo, N., Huang, D., Cambria, E., Wang, Z. (2025). Large Language Models for Logical Fallacy Detection. In: Yuan, S., Malliaros, F., Zheng, X. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2025*. Lecture Notes in Computer Science(), vol 15835. Springer, Singapore. https://doi.org/10.1007/978-981-96-8197-6_29
15. D. Huang and Z. Wang, "LLMs at the Edge: Performance and Efficiency Evaluation with Ollama on Diverse Hardware," *2025 International Joint Conference on Neural Networks (IJCNN)*, Rome, Italy, 2025, pp. 1-8, doi: 10.1109/IJCNN64981.2025.11228317.
16. Huang, D., Nguyen, T. S., Liausvia, F., & Wang, Z. (2025, April). RAP: A Metric for Balancing Repetition and Performance in Open-Source Large Language Models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 1479-1496).
17. Huang, D., Malwe, G., Varshney, R., Tse, A., & Wang, Z. (2025, March). Scalable Invoice Reconciliation for SMEs: Edge-Deployed LLMs in Multi-Agent Systems. In *2025 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI)* (pp. 1-6). IEEE.
18. D. Huang and Z. Wang, "Optimizing Chinese-to-English Translation Using Large Language Models, " *2025 IEEE Symposium on Computational Intelligence in Natural Language Processing and Social Media (CI-NLPSoMe Companion)*, Trondheim, Norway, 2025, pp. 1-7, doi: 10.1109/CI-NLPSoMe64976.2025.10970768.
19. D. Huang and Z. Wang, "Logical Reasoning with LLMs via Few-Shot Prompting and Fine-Tuning: A Case Study on Turtle Soup Puzzles," *2025 IEEE Symposium on Computational Intelligence in Natural Language Processing and Social Media (CI-NLPSoMe Companion)*, Trondheim, Norway, 2025, pp. 1-5, doi: 10.1109/CI-NLPSoMeCompanion65206.2025.10977921.
20. Wang, Z., Huang, D., Cui, J. et al. (2025). A review of Chinese sentiment analysis: subjects, methods, and trends. Artif Intell Rev 58, 75 (2025).
21. Huang, D., Fu, X., Yin, X., Pen, H., & Wang, Z. (2024). Automating maritime risk data collection and identification leveraging large language models. In *2024 IEEE International Conference on Data Mining Workshops (ICDMW)* (pp. 433–439). IEEE.
22. Huang, D., Hu, Z., & Wang, Z. (2024, June). Performance Analysis of Llama 2 Among Other LLMs. In *2024 IEEE Conference on Artificial Intelligence (CAI)* (pp. 1081-1085). IEEE.
23. Huang, D., & Wang, Z. (2024, April). Evaluation of Orca 2 Against Other LLMs for Retrieval Augmented Generation. In: Wang, Z., Tan, C.W. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2024*. Lecture Notes in Computer Science(), vol 14658. Springer, Singapore. https://doi.org/10.1007/978-981-97-2650-9_1
24. Huang, D., Samuel, S., Hyunh, Q. T., & Wang, Z. (2024, April). From Tweets to Token Sales: Assessing ICO Success Through Social Media Sentiments. In: Wang, Z., Tan, C.W. (eds) *Trends and Applications in Knowledge Discovery and Data Mining. PAKDD 2024*. Lecture Notes in Computer Science(), vol 14658. Springer, Singapore. https://doi.org/10.1007/978-981-97-2650-9_5

Theoretical Physics

25. Lin, W. B., Huang, D. H., Zhang, X., & Brandenberger, R. (2001). Nonthermal production of weakly interacting massive particles and the subgalactic structure of the universe. *Physical Review Letters*, *86*(6), 954.
26. Huang, D. H., Lin, W. B., & Zhang, X. M. (2000). Remark on approximation in the calculation of the primordial spectrum generated during inflation. *Physical Review D*, *62*(8), 087302.